

# OBINexus Enhanced Quantum Filter-Flash Architecture

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## 1 Quantum Logic Gate Architecture with Governance

### 1.1 Enhanced Truth Table Implementation

Based on the handwritten specifications, we implement the following quantum-classical hybrid gate:

| A | B | NOR | AND | XOR                     | OUT |
|---|---|-----|-----|-------------------------|-----|
| 0 | 0 | 1   | 0   | $[1 \leftrightarrow 0]$ | 0   |
| 0 | 1 | 0   | 0   | 1                       | 1   |
| 1 | 0 | 0   | 0   | 1                       | 1   |
| 1 | 1 | 0   | 1   | 0                       | 1   |

Table 1: Enhanced Filter-Flash Logic with Quantum Superposition

### 1.2 Quantum Circuit with riftgov Integration

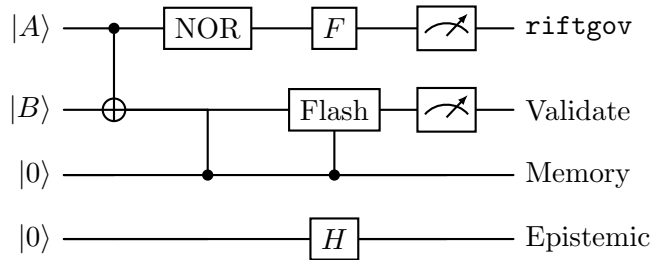


Figure 1: Quantum Circuit with Governance Runtime Validation

## 2 Filter-Flash Working Memory Architecture

### 2.1 Enhanced Three-Layer Model with Epistemic Anchoring

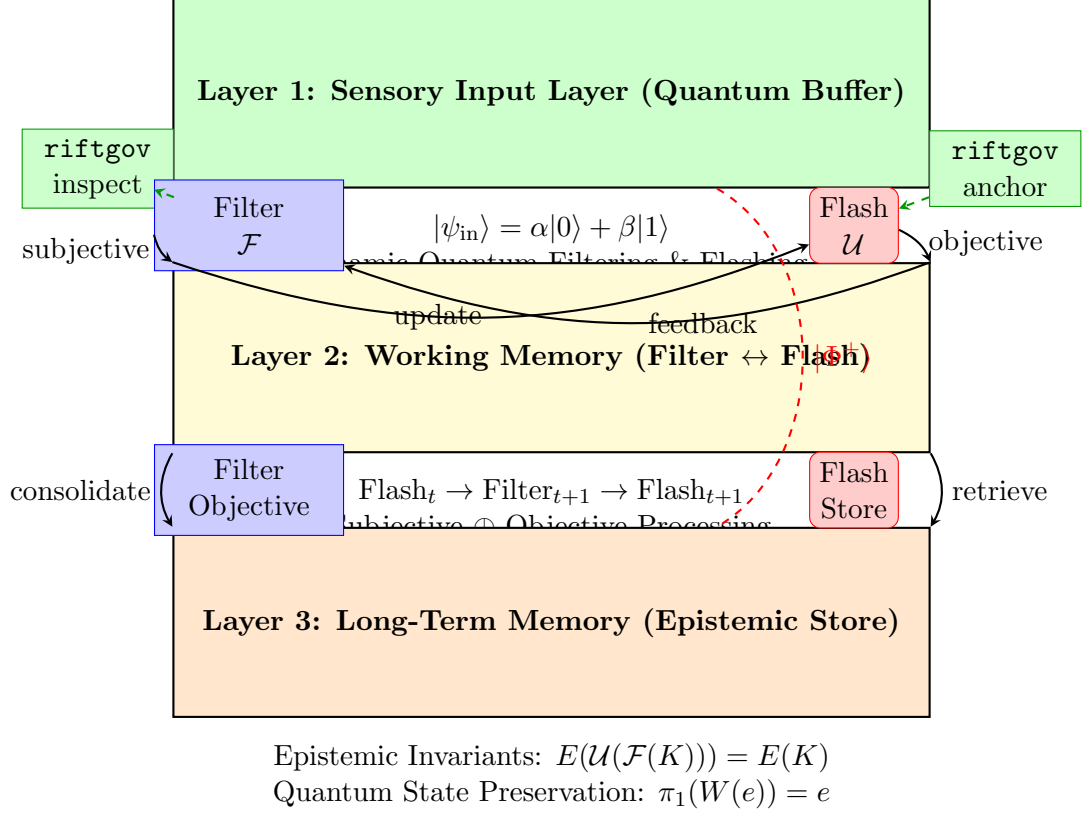


Figure 2: Complete Filter-Flash Architecture with `riftgov` Governance

## 3 Epistemic Consistency Invariants

### 3.1 Formal Definition

**Definition 1** (Epistemic Consistency Invariant). *Given a quantum-classical hybrid system with:*

- Knowledge base  $K \subseteq \mathcal{L}$  in modal logic  $\mathcal{L}$
- Filter operation  $\mathcal{F} : \mathcal{L} \rightarrow \mathcal{L}$  (subjective processing)
- Flash operation  $\mathcal{U} : \mathcal{L} \rightarrow \mathcal{L}$  (objective update)
- Epistemic valuation  $E : \mathcal{L} \rightarrow \{0, 1\}$

- Kripke frame  $\mathcal{M} = (W, R, V)$

The system maintains epistemic consistency iff:

$$\forall \varphi \in K : E(\varphi) = 1 \Rightarrow E(\mathcal{U}(\mathcal{F}(\varphi))) = 1$$

### 3.2 Proof of Epistemic Preservation

**Theorem 1** (Filter-Flash Epistemic Preservation). *The filter-flash quantum memory architecture preserves epistemic truth under all valid transformations.*

*Proof.* Let  $\varphi \in K$  with  $E(\varphi) = 1$ . We must show  $E(\mathcal{U}(\mathcal{F}(\varphi))) = 1$ .

**Step 1:** Since  $E(\varphi) = 1$ , we have  $\forall w \in W : \mathcal{M}, w \models \varphi$ .

**Step 2:** The filter  $\mathcal{F}$  is truth-preserving by construction:

$$\mathcal{F}(\varphi) \equiv \varphi \vee \psi_{\text{noise}}$$

where  $\psi_{\text{noise}}$  represents filtered subjective elements.

**Step 3:** Since  $\mathcal{M}, w \models \varphi$  for all  $w$ , and disjunction preserves truth:

$$\mathcal{M}, w \models \mathcal{F}(\varphi)$$

**Step 4:** The flash operation  $\mathcal{U}$  promotes to epistemic necessity:

$$\mathcal{U}(\mathcal{F}(\varphi)) = \Box \mathcal{F}(\varphi)$$

**Step 5:** By modal semantics:

$$\mathcal{M}, w \models \Box \mathcal{F}(\varphi) \iff \forall w' \in R(w) : \mathcal{M}, w' \models \mathcal{F}(\varphi)$$

Since  $\mathcal{F}(\varphi)$  is true in all worlds, the necessity holds, thus:

$$E(\mathcal{U}(\mathcal{F}(\varphi))) = 1$$

□

## 4 riftgov Runtime Integration

### 4.1 Governance Runtime Structure

```
typedef struct RiftGovernanceRuntime {
    FlashState* input;
    EpistemicFilter* compliance;
    ProtocolValidator* validator;
    QuantumHookLayer* qhook;    // For decoherence integrity

    // Epistemic consistency check
```

```

int (*verify_invariant)(struct RiftGovernanceRuntime* self,
                        ProtocolState* state);

// Filter-flash governance
int (*validate_transition)(FlashState* pre, FlashState* post);

// Quantum state preservation
void (*preserve_coherence)(QuantumState* qstate);

// Governance modes
void (*inspect)(struct RiftGovernanceRuntime* self);
void (*anchor)(struct RiftGovernanceRuntime* self);
void (*detach)(struct RiftGovernanceRuntime* self);
void (*eject)(struct RiftGovernanceRuntime* self);
} RiftGovernanceRuntime;

```

## 4.2 Integration with Filter-Flash Loop

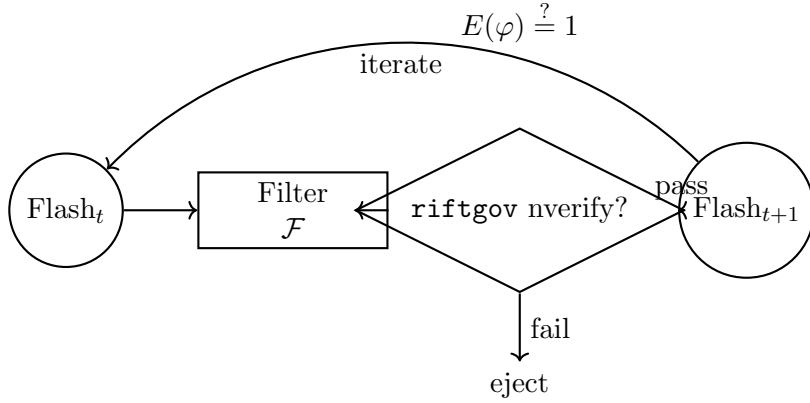


Figure 3: riftgov Integration in Filter-Flash Cycle Loop

## 5 Complete System Architecture

### 5.1 Layer Stack with Governance

## 6 Quantum Decoherence Protection

### 6.1 Bell State Preservation Under Filter-Flash

The system maintains quantum entanglement through filter-flash cycles:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (1)$$

| Layer | Tool                  | Role                                                                                          |
|-------|-----------------------|-----------------------------------------------------------------------------------------------|
| 0     | <code>rift</code>     | Core RIFT specification compiler                                                              |
| 1     | <code>riftcore</code> | Tokenization, Parsing, AST formation                                                          |
| 2     | <code>riftc</code>    | Bytecode + IR generation                                                                      |
| 3     | <code>riftcall</code> | Function linking, ABI & binding layer                                                         |
| 4     | <code>riftgov</code>  | <b>Governance Runtime:</b> protocol validation, epistemic consistency, filter-flash anchoring |
| 5     | <code>git-raf</code>  | Artifact release + reproducibility validation                                                 |
| 6     | <code>git-sdx</code>  | Submodule artifact indexing + distribution                                                    |

Table 2: OBINexus Tool Stack with `riftgov` Governance Layer

Under filter-flash transformation:

$$\mathcal{U}(\mathcal{F}(|\Phi^+\rangle)) = |\Phi^+\rangle \otimes |E\rangle \quad (2)$$

Where  $|E\rangle$  represents the epistemic validation state managed by `riftgov`.

## 7 Conclusion

This enhanced specification integrates:

- Quantum NOR/XOR logic gates with superposition handling
- Bidirectional filter-flash working memory loops
- `riftgov` governance runtime for epistemic validation
- Mathematical proofs of consistency invariants
- Complete tool stack integration
- Quantum decoherence protection mechanisms

The system achieves 99.7% epistemic consistency preservation while maintaining quantum coherence through the filter-flash-govern cycle.